

Let the imposed **surface** temperature drive be

$$T_s(t) = T_0 t^\tau.$$

The Marshak boundary condition is

$$\frac{4}{c} F_{\text{inc}}(t) = a_R T_s^4(t) + \frac{2}{c} F_s(t),$$

and for a bath temperature T_{bath} ,

$$F_{\text{inc}}(t) = \frac{a_R c}{4} T_{\text{bath}}^4(t).$$

Therefore,

$$a_R T_{\text{bath}}^4(t) = a_R T_s^4(t) + \frac{2}{c} F_s(t),$$

so

$$T_{\text{bath}}^4(t) = T_s^4(t) + \frac{2 F_s(t)}{a_R c}.$$

Since

$$a_R c = 4 \sigma_{\text{sb}},$$

this becomes

$$T_{\text{bath}}^4(t) = T_s^4(t) + \frac{F_s(t)}{2 \sigma_{\text{sb}}}.$$

From the self-similar radiation-flux representation,

$$F(\xi, t) = S(\xi) A^{a_S} B^{b_S} t^{c_S},$$

with the solver's flux powers

$$a_S = a + (q + 1)a_1 + na_3,$$

$$b_S = 1 + b + (q + 1)b_1 + nb_3,$$

$$c_S = c + (q + 1)c_1 + nc_3.$$

The dimensionless self-similar flux is

$$S(\xi) = -P^{n-1} V^q (V P' + k P V'),$$

At the surface/origin,

$$F_s(t) = F(0, t) = S_0 A^{a_s} B^{b_s} t^{c_s},$$

where

$$S_0 = -P_0^{n-1} V_0^q (V_0 P_0' + k P_0 V_0').$$

Substituting into the Marshak condition gives

$$T_{\text{bath}}^4(t) = T_0^4 t^{4\tau} + \frac{S_0 A^{a_s} B^{b_s}}{2\sigma_{\text{sb}}} t^{c_s}.$$

Factor out $T_0^4 t^{4\tau}$:

$$T_{\text{bath}}^4(t) = T_0^4 t^{4\tau} \left[1 + \frac{S_0 A^{a_s} B^{b_s}}{2\sigma_{\text{sb}} T_0^4} t^{c_s - 4\tau} \right].$$

Therefore,

$$T_{\text{bath}}(t) = T_0 t^\tau [1 + \mathcal{B}_{\text{bath}} t^{\eta_{\text{bath}}}]^{1/4}$$

with

$$\mathcal{B}_{\text{bath}} = \frac{S_0 A^{a_s} B^{b_s}}{2\sigma_{\text{sb}} T_0^4}$$

and

$$\eta_{\text{bath}} = c_s - 4\tau.$$

The correction term is dimensionless:

$$\mathcal{B}_{\text{bath}} t^{\eta_{\text{bath}}} = \frac{F_s(t)}{2\sigma_{\text{sb}} T_s^4(t)} = \frac{2F_s(t)}{a_{RC} T_s^4(t)}.$$

The dimensional constants are

$$A = T_0 [(\gamma - 1)f]^{1/\beta}$$

and

$$B = \frac{16\sigma_{\text{sb}} g}{3\beta [(\gamma - 1)f]^n}.$$

Here B is the **diffusion prefactor** used by the self-similar solution, not $\mathcal{B}_{\text{bath}}$.

Thus the bath coefficient may be written explicitly as

$$\mathcal{B}_{\text{bath}} = \frac{S_0}{2\sigma_{\text{sb}}T_0^4} \left[T_0 ((\gamma - 1)f)^{1/\beta} \right]^{a_S} \left[\frac{16\sigma_{\text{sb}}g}{3\beta ((\gamma - 1)f)^n} \right]^{b_S}.$$

With

$$n = \frac{4 + \alpha}{\beta}, \quad k = 1 - \mu, \quad q = kn + \lambda - 1,$$

the pure-power forms are

$$a_S = \frac{3\beta\lambda + 2\beta + 2\alpha + 8 - \mu(3\alpha + 12)}{4 + 2\lambda - 4\mu}$$

$$b_S = \frac{2 - 3\mu}{4 + 2\lambda - 4\mu}$$

$$c_S = \frac{\tau [3\beta\lambda + 2\beta + 2\alpha + 8 - \mu(3\alpha + 12)] + 3\mu - 2}{4 + 2\lambda - 4\mu}$$

and therefore

$$\eta_{\text{bath}} = \frac{\tau [3\beta\lambda + 2\beta + 2\alpha - 8 - 8\lambda + 4\mu - 3\alpha\mu] + 3\mu - 2}{4 + 2\lambda - 4\mu}.$$

So the final result is

$$T_{\text{bath}}(t) = T_0 t^\tau \left[1 + \frac{S_0}{2\sigma_{\text{sb}}T_0^4} \left[T_0 ((\gamma - 1)f)^{1/\beta} \right]^{a_S} \left[\frac{16\sigma_{\text{sb}}g}{3\beta ((\gamma - 1)f)^n} \right]^{b_S} t^{\eta_{\text{bath}}} \right]^{1/4}.$$